

Liability Hedging

Assets chosen in same way as liabilities

Eg: Immunisation

- ↳ match by term to reduce interest rate sensitivity

CF matching using bonds

Disadvantages **CADMIT**

- only partial hedge if asset cover < 100%
- reinvestment risk → duration mismatch
- gaps in maturities = reinvest or disinvest bond prior to maturity → imperfect hedge
- small degree of credit risk in government bonds → liabilities may not reject
- tax status of government bonds may worsen → Δ insufficient to meet L
- market to market risk

CF matching using bonds + swaps

- ↳ Synthetic portfolio management

FLAT IN

- longer maturity
- greater liquidity
- lower transaction costs
- allow hedge without full asset cover
- flexibility in schedule of payments → bespoke

DISADV

- direct swap needs Isbn documentation $\begin{cases} \text{expensive} \\ \text{time consuming} \end{cases}$
- collateralisation movement
- bespoke: closing out is complex
- counterparty risk → default
- investment risk → meeting payments
- basis risk leading to mark to market risk
 - ↳ swap int rate curve moves differently to govern bond int rate curve

→ If assumptions are borne out; then covered

CHAPTER 20:
ACTUARIAL
TECHNIQUES

Liability Driven Investment

Investment decision where asset allocation is determined in whole/part relative to a set of liabilities

closely match:

- interest rate sensitivity
- inflation-linkage
- shape of liabilities

Approaches:

- **swap portfolios**
- long-duration bond management

hedge exposure to following risks

① Interest Rate Risk

- ↑ int rate = ↓ PV_L
- greater length of time to CF = more sensitive to Δ in int rate
- reduce risk by matching duration
 - ↳ fixed rate bond
 - ↳ interest rate swap

② Inflation Risk

- sensitive to changes in expectations on inflation
 - ↳ inflation-linked bonds
 - ↳ inflation swaps

PV01 → measure sensitivity of value of L to Δ in int rate

↳ Δ in PV_L due to 1 basis point move in int rate

DV01 → measures same Δ when Dollar value is used to quote L